

The 2-Center Problem in Chordal Graphs

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Abstract

In a graph G , the p -center problem is to identify p vertices of G for locating some kind of facilities such that the maximum among the distances from all vertices to their nearest facility is minimized. In this paper, we propose an $O(n)$ -time algorithm for computing a 2-center in a chordal graph with maximum degree at most three, where a graph is chordal if every induced cycle of length at least four has a chord, and n is the number of vertices in the graph.

Keywords: Chordal graphs; 2-center

1 Introduction

Facility location problems are widely investigated in the fields of transportation, operations research, and theoretical computer sciences [1, 7]. The p -center problem is one of the well-investigated problem in the literature [4]. In the p -center problem, one is asked to deploy p facilities in a given graph so as to minimize the longest distance between a set of n demand points and the p facilities. The p -center problem for arbitrary p in general graphs is one of the well known NP-hard problems in graph theory [8]. A graph is called a chordal graph (which is an edge joining two nodes that are not adjacent in the cycle of length greater than three) if and only if one can repeatedly find a simplicial vertex and delete it from the graph, until no vertex is left [6]. In [3] the inequality $2r(G) - 2 \leq d(G) \leq 2r(G)$ was proven and the characteristic of center on chordal graph, see [2]. The class of chordal graphs contains trees, split graphs, ptolemaic graphs, k -trees, interval graphs and strongly chordal graphs. The chordal graphs are a subclass of perfect graphs. Other superclasses of chordal graphs include the weakly chordal graphs, the odd-hole-free graphs, and the even-hole-free graphs. In fact, chordal graphs are

exactly the graphs that are both odd-hole-free and even-hole-free.

On chordal graphs, the p -center problem for $p = 1$, can be solved in $O(m)$ time, where m is the number of edges in the given graph [5]. In this paper, we consider the p -center problem on chordal graphs with $\Delta(G) \leq 3$ and $p = 2$ and present an $O(n)$ -time algorithm, where n is the number of vertices in the given graph.

The remainder of this paper is organized as follows. In Section 2, we formally define the problem and the notation used in this paper. In Section 3, we present a linear time algorithm. Finally, concluding remarks are given in Section 4.

2 Preliminaries

In this section, we define the problems and notations used in this article. Readers can refer to West [9] for terminology not defined here. All graphs considered in this paper are connected, undirected, loopless, and without multiple edges. Given a graph $G = (V, E)$, where V and E are the vertex set and the edge set, respectively, n is the cardinality of V and m is the cardinality of E . For succinctness, the vertex set of a subgraph G' of G is also referred to as G' . The subgraph of G induced by $U \subseteq V$ is denoted by $G[U]$. The *degree* of a vertex v is the number of edges incident with v and is denoted by $\deg(v)$. The *maximum degree* of G is defined to be $\Delta(G) = \max_{v \in V} \deg(v)$. The *neighborhood* of a vertex u is defined to be $N_G(u) = \{v : v \in V \text{ and } uv \in E\}$, and the neighborhood of a vertex subset V' is defined as $N_G(V') = \bigcup_{v \in V'} N_G(v) - V'$.

In a graph $G = (V, E)$, a path from a vertex u to a vertex v is denoted by $\mathcal{P}[u, v]$. The *length* of a path is the number of edges inside the path. Similarly, the length of a cycle is defined to be the number of edges contained in the cycle, and a k -cycle is a cycle of length k . The *distance* $d(u, v)$ from a vertex u to a vertex v is the length of a shortest path from u to v , and the *distance from a vertex*

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v to a set of vertices U is defined to be $d(v, U) = \min_{u \in U} d(v, u)$. The *eccentricity* of a vertex v is defined as $e(v) = \max_{u \in V} d(u, v)$, and the eccentricity of a vertex v with respect to a subgraph G' of G is defined as $\text{ecc}(v, G') = \max_{u \in G'} d(u, v)$. A *diameter* of a graph $G = (V, E)$ is defined to be a shortest path $\mathcal{P}[u, u']$, where $d(u, u') = \max_{v \in V} e(v)$, and a diameter of a subgraph G' of G is denoted by $D(G')$. The *radius* of a graph $G = (V, E)$ is defined as $r(G) = \min_{v \in V} e(v)$, and the length of a diameter of G is denoted by $d(G)$.

The 2-center problem is formally defined as follows.

Definition 1 (the 2-center problem). *Given a graph $G = (V, E)$, the 2-center problem asks for a pair of vertices $\{c_1, c_2\}$ which minimizes the objective function*

$$e(u_1, u_2) = \max_{v \in V} d(v, \{u_1, u_2\}).$$

The vertex pair $\{c_1, c_2\}$ is called a *2-center* of G , and we denote $e(c_1, c_2)$ by $r_2(G)$.

In the following, some properties of a chordal graph and a 2-center in general graphs are given in Lemmas 1, 2, and 3.

Lemma 1. *Let $G = (V, E)$ be a connected graph. There is a partition (V_1, V_2) of V , which satisfies*

- $G[V_i]$ is connected, for $i = 1, 2$.
- Let c_i be a central vertex of $G[V_i]$, for $i = 1, 2$. The pair $\{c_1, c_2\}$ is a 2-center of G .

Proof. Let $\{c'_1, c'_2\}$ be a 2-center of G . We define the partition as follows. Let $V_1 = \{v : d(v, c'_1) \leq d(v, c'_2)\}$, and $V_2 = V - V_1$.

claim: $G[V_i]$ is connected, for $i = 1, 2$.

Consider the case where $i = 1$. Suppose to the contrary that $G[V_1]$ is not connected. There are at least two components, C_1 and C_2 , in $G[V_1]$. Without loss of generality, we assume that $c'_1 \in C_1$. Let $x \in C_2$ satisfying $d(x, c'_1) = d(y, c'_1) + 1$, where $y \in V_2$. Note that x is well defined since in a shortest path between c'_1 and some vertex in C_2 , there is always a pair of consecutive vertices (u, v) , that belong to C_2 and V_2 , respectively. Apparently, $d(u, c'_1) = d(v, c'_1) + 1$, i.e., $(u, v) = (x, y)$. By definition, we have $d(y, c'_2) < d(y, c'_1)$ and $d(x, c'_2) \leq d(y, c'_2) + 1$. Thus,

$$d(x, c'_2) \leq d(y, c'_2) + 1 < d(y, c'_1) + 1 = d(x, c'_1),$$

which contradicts that $d(x, c'_1) \leq d(x, c'_2)$.

For $i = 2$, the argument is similar and therefore omitted.

claim: The pair $\{c_1, c_2\}$ is a 2-center of G .

To prove this claim, it suffices to show that $\min\{d(x, c_1), d(x, c_2)\} \leq e(c'_1, c'_2)$ for all $x \in V$. Because of symmetry, we show only the case where $x \in V_1$. Since $x \in V_1$, by the definition of eccentricity, we have $d(x, c_1) \leq \text{ecc}(c_1, V_1)$. Moreover, since c_1 is a central vertex of $G[V_1]$, we have $\text{ecc}(c_1, V_1) \leq \text{ecc}(c'_1, V_1)$. Thus,

$$\begin{aligned} & \min\{d(x, c_1), d(x, c_2)\} \\ & \leq d(x, c_1) \leq \text{ecc}(c_1, V_1) \leq \text{ecc}(c'_1, V_1) \leq e(c'_1, c'_2), \end{aligned}$$

which completes the proof. $\square$

For a graph $G = (V, E)$, a partition (V_1, V_2) of V is called an *optimal partition* if (V_1, V_2) satisfies the two conditions in Lemma 1.

Lemma 2. *Let G be a graph and D be a diameter of G . There is an optimal partition (V_1, V_2) satisfying $V_i \cap D \neq \emptyset$, for $i = 1, 2$.*

Proof. Let $\{c_1, c_2\}$ be a 2-center of G . We have $e(c_1, c_2) \leq r(G)$. If $e(c_1, c_2) < r(G)$, the statement holds trivially. If $e(c_1, c_2) = r(G)$, we can define an optimal partition as $(\{v_1\}, V - \{v_1\})$, where $D = (v_1, v_2, \dots, v_k)$. To show that $(\{v_1\}, V - \{v_1\})$ is an optimal partition, it suffices to show that $V - \{v_1\}$ induces a connected subgraph. Since v_1 is an end vertex of D , it is not a cut vertex of G . Thus, $V - \{v_1\}$ induces a connected subgraph, and the lemma is proven. $\square$

Lemma 3. *In a chordal graph G , an edge e is contained in a cycle if and only if e is contained in a 3-cycle.*

Proof. The sufficiency is trivial. For necessity, consider a minimal cycle χ that contains e . If χ is not a 3-cycle, there must be a chord in χ , which results in a shorter cycle that contains e . According to the minimal principle, there is contradiction. $\square$

3 The 2-center on chordal graphs with $\Delta(G) \leq 3$

In this section, we propose a linear time algorithm which computes a 2-center of a chordal graphs G with $\Delta(G) \leq 3$. A sketch of the proposed algorithm is given below.

- Step 1 Find a diameter D of G .
- Step 2 Determine an optimal partition (V_1, V_2) that divides D .
- Step 3 Compute a central vertex c_i of $G[V_i]$, for $i = 1, 2$.

By Lemmas 1 and 2, the pair $\{c_1, c_2\}$ found in Step 3 is a 2-center of G . Steps 1 and 3 can be done by the linear time algorithm given in [5]. Therefore, we focus on the implementation of Step 2. We shall show that there is a tree-like structure of a chordal graph with maximum degree less than or equal to three in Section 3.1. Based on the structure, a linear time algorithm for determining an optimal partition is developed in Section 3.2.

3.1 The structure of a chordal graph with maximum degree less than or equal to three

Now we present structural properties of the chordal graph with $\Delta(G) \leq 3$.

Lemma 4. *In chordal graph G , an edge uv is a cut edge if and only if $N(u) \cap N(v) = \emptyset$.*

Proof. For necessity, the statement holds since otherwise there is a path $\mathcal{P}[u, v]$ in $G - uv$. For sufficiency, we prove the contrapositive. If uv is not a cut edge, there is a path $\mathcal{P}[u, v]$ in $G - uv$. Together with uv , there is a cycle containing uv in G , and therefore, by Lemma 3, a 3-cycle containing uv in G . Let (u, v, x) be the 3-cycle. We have $x \in N(u) \cap N(v) \neq \emptyset$. $\square$

Lemma 5. *In a chordal graph G with $\Delta(G) \leq 3$, let $D = (v_1, v_2, \dots, v_k)$ be a diameter. $G = K_4$ if and only if $\deg(v_1) = 3$.*

Proof. Necessity is trivial. For sufficiency, suppose to the contrary that $G \neq K_4$. Since $\deg(v_1) = 3$, $|V(G)| > 3$. Furthermore, since $\Delta(G) \leq 3$, $G \neq K_n$ for $n > 4$. Therefore, it suffices to consider the case where $|V(G)| \geq 4$ and $k \geq 3$.

Since v_1 is an end vertex of D , we have $|N(v_1) \cap D| = 1$. Let $N(v_1) - D = \{x_1, x_2\}$. We have that $N(x_i) \cap D - \{v_1\} \neq \emptyset$, for $i = 1, 2$, since otherwise D can be extended to be a longer path. Moreover,

$N(x_i) - D \subseteq \{v_2, v_3\}$. Since $\Delta(G) \leq 3$, we have $|N(v_i) - D| \leq 1$ for $2 \leq i \leq k - 1$. It follows that $N(x_i) \cap D = \{v_1, v_3\}$ for some $i = 1, 2$. Without loss of generality, let $i = 1$. There is a 4-cycle (x_1, v_1, v_2, v_3) without any chord, which is a contradiction. $\square$

Now we prove that the optimal edge cut is a cut edge or a pair of edges. Alternatively, we denote an edge cut S by $[U, \bar{U}]$, where each edge $uv \in S$ satisfies $u \in U$ and $v \in \bar{U}$.

Lemma 6. *For a chordal graph G with $\Delta(G) \leq 3$ and $G \neq K_4$, let D be a diameter of G and $\mathcal{S}$ be the set of optimal edge cuts that divide D . If $S = \arg \min_{S' \in \mathcal{S}} |S'|$, then $|S| \leq 2$.*

Proof. Suppose to the contrary that $|S| \geq 3$. Let $uv \in S \cap D$. By Lemma 4, uv is contained in a 3-cycle (u, v, w) . Without loss of generality, we assume that $uw \in S$. Choose among $S - \{uv, uw\}$ an edge which constitutes a shortest cycle with uv or uw . It is noted that such a cycle is a 3-cycle since otherwise there has to be a chord that reduces the cycle length. Without loss of generality, let (u, w, x) be the 3-cycle. There are two possible cases: (i) $ux \in S$ and (ii) $wx \in S$. Let $S = [U, \bar{U}]$ and $u \in U$.

case (i): $ux \in S$.

In this case, $\deg(u) = \deg(w) = 3$. Since $G \neq K_4$, u is not an end vertex of D . Furthermore, $uw \notin D$ since $vw \in E(G)$. It follows that (v, u, x) is a subpath of D . By replacing (v, u, x) with (v, w, x) , we obtain another diameter contained in $\bar{U}$, which results in $r_2(G) = r(G)$. According to the argument in Lemma 2 and that $G \neq K_4$, we have $[y, V(G) - y] \in \mathcal{S}$ and $|[y, V(G) - y]| \leq 2 < |S|$, where y is the end vertex of D .

case (ii): $wx \in S$.

Let $U_w = U \cup \{w\}$. It suffices to show that $[U_w, \bar{U}_w] \in \mathcal{S}$ and $|[U_w, \bar{U}_w]| < |S|$. Since $|N(w) \cap U| = 2$ and $|N(w) \cap \bar{U}| = 1$, we have $|[U_w, \bar{U}_w]| = |S| - 1$. To show that $[U_w, \bar{U}_w] \in \mathcal{S}$, we claim that $r(U_w) = r(U)$. If $c_1 = u$, the claim holds trivially. Otherwise, it follows that

$$\begin{aligned} r(U_w) &\leq \text{ecc}(c_1, U_w) \\ &= \max\{r(U), d(c_1, w)\} \\ &= \max\{r(U), d(c_1, x) + 1\} \\ &= \max\{r(U), d(c_1, u)\} \\ &= r(U). \end{aligned}$$

Together with that $uv \in [U_w, \bar{U}_w] \cap D$ and $r(\bar{U}_w) \leq r(\bar{U})$, we have $[U_w, \bar{U}_w] \in \mathcal{S}$. $\square$

Let $N(D) = \bigcup_{i=1}^3 \Gamma_i$, where $\Gamma_i = \{v : v \in V - D \text{ and } |N(v) \cap D| = i\}$. Because G is chordal, if $v \in \Gamma_2$, then $N(v) \cap D = \{v_i, v_{i+1}\}$ for some $1 \leq i < k$. Similarly, if $v \in \Gamma_3$, then $N(v) \cap D = \{v_i, v_{i+1}, v_{i+2}\}$ for some $1 \leq i \leq k - 2$.

Corollary 1. *Let G be a chordal graph with $\Delta(G) \leq 3$ and $D = (v_1, v_2, \dots, v_k)$ be a diameter of G with $k > 2$. If S is a minimum optimal edge cut that divides D and $|S| = 2$, then $S = \{v_i v_{i+1}, v_i u\}$ or $\{v_i v_{i+1}, v_{i+1} u\}$ for some i , where $1 \leq i < k$ and $u \in \Gamma_3$.*

Proof. Since S divides D , $v_i v_{i+1} \in S$ for some i , $1 \leq i < k$. Moreover, since $|S| = 2$ and S is minimum, by Lemma 3, there is a vertex $u \in N(v_i) \cap N(v_{i+1})$. Apparently, $u \in \Gamma_i$ for $i = 2$ or 3 , and therefore $S = \{v_i v_{i+1}, v_i u\}$ or $\{v_i v_{i+1}, v_{i+1} u\}$. Suppose that $u \in \Gamma_2$. Since $k > 2$ and $\Delta(G) \leq 3$, there is a cut edge $v_{i-1} v_i$ or $v_{i+1} v_{i+2}$. We can replace S with either $v_{i-1} v_i$ or $v_{i+1} v_{i+2}$, which results in a contradiction. $\square$

3.2 Determining an optimal partition

For a diameter $D = (v_1, v_2, \dots, v_k)$, based on Lemma 6 and Corollary 1, a *candidate cut* is an edge cut S , where

$$S = \begin{cases} \{v_i v_{i+1}\}, & v_i v_{i+1} \text{ is a cut edge;} \\ \{v_i v_{i+1}, v_i u\}, & u \in \Gamma_3; \\ \{v_i v_{i+1}, v_{i+1} u\}, & u \in \Gamma_3. \end{cases}$$

For indexed objects $O = \{O_1, O_2, \dots, O_k\}$, we define the *index function* $\text{ind} : O \mapsto \{1, 2, \dots, k\}$ as $\text{ind}(O_i) = i$, for $1 \leq i \leq k$. A *stem* of D is defined to be a component of G after removing all candidate cuts. All the stems can be ordered as $(C_1, C_2, \dots, C_s)$, where $\text{ind}(C_i) < \text{ind}(C_j)$ implies $\exists v \in C_i \cap D, \text{ind}(v) < \text{ind}(u)$ for all $u \in C_j \cap D$. The *pivot* σ_i of C_i is defined to be a vertex u such that $u \in C_i \cap D$ and $\text{ind}(u) = \max_{v \in C_i \cap D} \text{ind}(v)$. The *branch* C_{l_i} is defined to be $G \left[\bigcup_{j \leq i} C_j \right]$. An illustration is given in Figure 1.

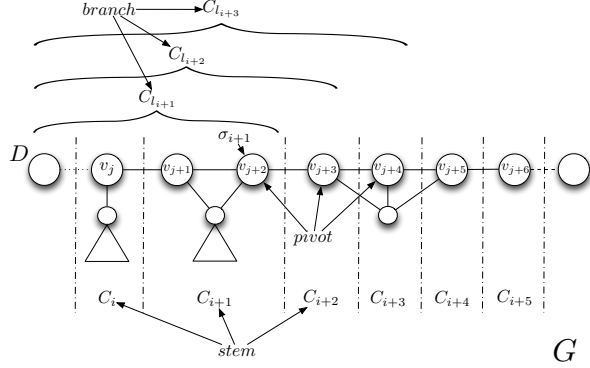


Fig. 1. Stems, pivots, and branches of a chordal graph G with respect to a diameter D , where $\Delta(G) \leq 3$.

Lemma 7. *For a diameter $D = (v_1, v_2, \dots, v_k)$ with $k > 2$ and a stem C_i , we have $|C_i \cap D| \leq 2$. In particular, if $|C_i \cap D| = 2$, then $C_i \cap D = \{v_x, v_{x+1}\}$ and $\text{ecc}(v_x, C_i) = \text{ecc}(v_{x+1}, C_i) = \text{ecc}(\sigma_i, C_i)$ for some $x \in \{1, 2, \dots, k\}$.*

Proof. Suppose to the contrary that $|C_i \cap D| > 2$. Let $C_i \cap D = \{v_{i_1}, v_{i_2}, \dots, v_{i_s}\}$. Without loss of generality, we assume that $i_j < i_{j+1}$ for $1 \leq j < s$. It follows that $i_{j+1} - i_j = 1$ since otherwise there is an edge cut in C_i . By Lemma 4, $\bigcap_j N(v_{i_j}) \neq \emptyset$. Together with that $\Delta(G) \leq 3$, we have $|C_i \cap D| = 3$. However, such a component contains a candidate cut, which is a contradiction.

For $|C_i \cap D| = 2$, there is exactly one possibility where $C_i \cap D = \{v_x, v_{x+1}\}$ for some $x \in \{1, 2, \dots, k\}$ and $N(v_x) \cap N(v_{x+1}) = \{u\} \subseteq \Gamma_2$. It follows that $\sigma_i = v_{x+1}$. If $\text{ecc}(v_x, C_i) = 1$, we have $\text{ecc}(v_{x+1}, C_i) = 1$. Otherwise, we have

$$\text{ecc}(v_x, C_i) = \min\{\text{ecc}(v_{x+1}, C_i), \text{ecc}(u, C_i)\} + 1,$$

and

$$\text{ecc}(v_{x+1}, C_i) = \min\{\text{ecc}(v_x, C_i), \text{ecc}(u, C_i)\} + 1,$$

which implies

$$\text{ecc}(v_x, C_i) = \text{ecc}(v_{x+1}, C_i) = \text{ecc}(\sigma_i, C_i).$$

$\square$

We determine an optimal edge cut by computing the radii of U and $\bar{U}$ with respect to all candidate cuts $[U, \bar{U}]$. Apparently, U is a branch and therefore $U = C_{l_i}$ for some i . We show in Lemma 8 that the length of a diameter $d(U)$ of U can be computed recursively using $d(C_i)$, $d(C_{l_{i-1}})$, $\text{ecc}(\sigma_{i-1}, C_{l_{i-1}})$, and $\text{ecc}(\sigma_i, C_i)$.

Lemma 8. For a chordal graph G with $\Delta(G) \leq 3$ and a diameter D of G , we have

$$d(C_{l_i}) = \begin{cases} d(C_1), & \text{if } i = 1; \\ \max\{d(C_i), d(C_{l_{i-1}}) + 1 + \text{ecc}(\sigma_{i-1}, C_{l_{i-1}})\}, & \text{if } i > 1 \text{ and } \sigma_i \in N(\Gamma_3); \\ \max\{d(C_i), d(C_{l_{i-1}}) + 1 + \text{ecc}(\sigma_{i-1}, C_{l_{i-1}}) + \text{ecc}(\sigma_i, C_i)\}, & \text{otherwise,} \end{cases}$$

and

$$\text{ecc}(\sigma_i, C_{l_i}) = \begin{cases} \text{ecc}(\sigma_1, C_1), & \text{if } i = 1; \\ \max\{\text{ecc}(\sigma_i, C_i), \text{ecc}(\sigma_{i-1}, C_{l_{i-1}}) + 2\}, & \text{if } i > 1 \text{ and } \sigma_i \in N(\Gamma_2); \\ \max\{\text{ecc}(\sigma_i, C_i), \text{ecc}(\sigma_{i-1}, C_{l_{i-1}}) + 1\}, & \text{otherwise,} \end{cases}$$

where C_{l_i} is a branch, C_i is a stem, and σ_i is a pivot of C_i .

Proof. For $i = 1$, by definition, $C_{l_1} = C_1$. Thus, that $d(C_{l_1}) = d(C_1)$ and $\text{ecc}(\sigma_1, C_{l_1}) = \text{ecc}(\sigma_1, C_1)$ follows the directly.

To compute $d(C_{l_i})$ for $i > 1$, since $C_{l_i} = C_{l_{i-1}} \cup C_i$, there are three possible cases: (i) $D(C_{l_i}) \subseteq C_{l_{i-1}}$; (ii) $D(C_{l_i}) \subseteq C_i$; (iii) $D(C_{l_i}) \cap C_{l_{i-1}} \neq \emptyset$ and $D(C_{l_i}) \cap C_i \neq \emptyset$. Apparently, for cases (i) and (ii), $d(C_{l_i}) = d(C_{l_{i-1}})$ and $d(C_i)$, respectively. For case (iii), by definition, all paths from a vertex in $C_{l_{i-1}}$ to a vertex in C_i contain an edge in a candidate cut S .

- If $S = \{v_x v_{x+1}\}$, it follows that $d(C_{l_i}) = 1 + \text{ecc}(v_x, C_{l_{i-1}}) + \text{ecc}(v_{x+1}, C_i)$. By Lemma 7, we have $\text{ecc}(v_{x+1}, C_i) = \text{ecc}(\sigma_i, C_i)$. Together with that $v_x = \sigma_{i-1}$, $d(C_{l_i}) = 1 + \text{ecc}(\sigma_{i-1}, C_{l_{i-1}}) + \text{ecc}(\sigma_i, C_i)$.
- If $S = \{v_x v_{x+1}, v_x u\}$ or $\{v_x v_{x+1}, v_{x+1} u\}$, $u \in \Gamma_3$, it follows that v_{x+1} or u is an end vertex of $D(C_{l_i})$, which implies that $d(C_{l_i}) = 1 + \text{ecc}(\sigma_{i-1}, C_{l_{i-1}})$.

Similarly, to compute $\text{ecc}(\sigma_i, C_{l_i})$, we consider the following two cases: (i) the farthest vertex to σ_i belongs to C_i ; (ii) the farthest vertex to σ_i belongs to $C_{l_{i-1}}$. For case (i), that $\text{ecc}(\sigma_i, C_{l_i}) = \text{ecc}(\sigma_i, C_i)$ follows directly. For case (ii), if $C_i \cap$

$D = \{\sigma_i\}$, we have that σ_i is adjacent to σ_{i-1} , and therefore $\text{ecc}(\sigma_i, C_{l_i}) = \text{ecc}(\sigma_{i-1}, C_{l_{i-1}}) + 1$. Otherwise, $d(\sigma_{i-1}, \sigma_i) = 2$. Thus, $\text{ecc}(\sigma_i, C_{l_i}) = \text{ecc}(\sigma_{i-1}, C_{l_{i-1}}) + 2$. $\square$

Once we have the length of a diameter of U , we can compute the radius of U by Corollary 2, which is based on Theorem 1.

Theorem 1 (See [3]). $2r(G) - 2 \leq d(G) \leq 2r(G)$ for any connected chordal graph G . Moreover, if $2r(G) - 2 = d(G)$, then G has a 3-sun as an induced subgraph.

Corollary 2. Let G be a chordal graph with $\Delta(G) \leq 3$. We have $r(G) = \left\lceil \frac{d(G)}{2} \right\rceil$.

Proof. Since $\Delta(G) \leq 3$, G has no 3-sun as an induced subgraph. Therefore, by Theorem 1, we have $2r(G) - 1 \leq d(G) \leq 2r(G)$. It follows that

$$\left\lceil \frac{d(G)}{2} \right\rceil \leq r(G) \leq \left\lfloor \frac{d(G) + 1}{2} \right\rfloor = \left\lceil \frac{d(G)}{2} \right\rceil.$$

$\square$

The radius of $\bar{U}$ can be computed symmetrically. By choosing the edge cut which results in a minimax radii among all candidate cuts, we can determine an optimal edge cut.

3.3 A linear time algorithm

A detailed description of computing a 2-center of a chordal graph G with $\Delta(G) \leq 3$ is given in CHORDAL-2CENTER(G).

PARTITION(G, D)

- 1 $(C_1, C_2, \dots, C_s) \leftarrow$ stems of G w.r.t. D
- 2 **for** $i \leftarrow 1$ **to** s
- 3 **do** Compute $d(C_i)$ and $\text{ecc}(\sigma_i, C_i)$
- 4 **for** $i \leftarrow 1$ **to** $s - 1$
- 5 **do** Compute $d(C_{l_i})$ and $\text{ecc}(\sigma_i, C_{l_i})$
- 6 Repeat Lines 1 to 5 by replacing $(C_1, C_2, \dots, C_s)$ with $(C'_1, C'_2, \dots, C'_s)$, where $C'_i = C_{s-i+1}$.
- 7 $j \leftarrow \arg \min_{1 \leq i < s} \max \left\{ \lceil d(C_{l_i})/2 \rceil, \lceil d(C'_{l_{s-i}})/2 \rceil \right\}$
- 8 **return** $(C_{l_j}, C'_{l_{s-j}})$

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CHORDAL-2CENTER( $G$ )
1   $D \leftarrow$  a diameter of  $G$ 
2  if  $|D| \leq 2$ 
3      then return arbitrary two vertices of  $G$ 
4  else
5       $(U, U') \leftarrow$  PARTITION( $G, D$ )
6       $c_1 \leftarrow$  a central vertex of  $U$ 
7       $c_2 \leftarrow$  a central vertex of  $U'$ 
8      return  $\{c_1, c_2\}$ 
    
```

The correctness of CHORDAL-2CENTER(G) follows from the results given in Sections 3.1 and 3.2. For the time complexity, the analysis is given in Theorem 2.

Theorem 2. *The 2-center of a chordal graph G with $\Delta(G) \leq 3$ can be computed in $O(n)$ time.*

Proof. It is noted that since $\Delta(G) \leq 3$, both n and m are of $O(n)$. In CHORDAL-2CENTER, except Line 5, all the other steps can be done in $O(n)$ time [5]. As for PARTITION, by Lemma 4, Corollary 1, and that $\Delta(G) \leq 3$, all the candidate cuts can be identified in $O(n)$ time. Therefore, Line 1 can be done in $O(n)$ time. Lines 2 and 3 can be accomplished by using the method in [5] (for $d(C_i)$) and breadth first search (for $\text{ecc}(\sigma_i, C_i)$), which results in time complexity $O(n)$. By Lemma 8, $d(C_{l_i})$ and $\text{ecc}(\sigma_i, C_{l_i})$ can be computed in constant time, given $d(C_{l_{i-1}})$, $d(C_i)$, $\text{ecc}(\sigma_i, C_{l_{i-1}})$, and $\text{ecc}(\sigma_i, C_i)$. Thus, Lines 4 and 5 can be done in $O(n)$ time. Since $s \leq n$, Line 7 can also be done in $O(n)$ time. As a result, the overall time complexity for CHORDAL-2CENTER is $O(n)$. $\square$

4 Concluding remarks

In this paper, we consider the 2-center problem on a chordal graph G with $\Delta(G) \leq 3$ and propose an $O(n)$ -time algorithm. For arbitrary chordal graphs, the 2-center problem can be solved by an $O(n^3)$ time naive method, in which the objective function is computed for all pairs of vertices. We would like to develop a more efficient al-

gorithm. Furthermore, the backup 2-center problem is that we are interested in, and more investigation will be conducted in the future.

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